

Assignment 1

Implementation of routing for the project

Objective: To design a routing system that meets the requirements of your project.

Description:

Each team develops a routing system for their project. It is necessary to create a set of routes that allow interaction with the server via CRUD data operations.

PS: DO NOT WRITE CRUD OPERATIONS, HERE ONLY DESCRIBES WHY WE NEED ROUTING.

Functional Requirements:

Basic Routes:

Implement routes for CRUD operations.

Use dynamic route parameters (e.g. /:id) where appropriate.

Middleware:

Add middleware to validate request data (e.g. ensure all required fields are specified in a POST request).

Modular Structure:

Organize your code using separate files for routes, controllers, and models (if applicable).

Route Testing:

Test each route using Postman or curl to make sure it works.

General requirements:

1. Define routes:

Analyze your project and define the required routes.

Create at least 3 routes covering different types of requests:

- GET to retrieve data.
- POST to create new resources.
- PUT or PATCH to update existing resources.
- DELETE to delete resources.

2. Examples of routes for different projects:

For the blog platform:

GET /posts - Get a list of all posts.

POST /posts - Create a new post.

GET /posts/:id - Get a post by ID.

PUT /posts/:id - Update a post by ID.

DELETE /posts/:id - Delete a post by ID.

For the e-commerce application:

GET /products - Get a list of all products.

GET /products/:id - Get product information by ID.

POST /cart - Add a product to the cart.

PUT /cart/:id - Change the quantity of a product in the cart.

DELETE /cart/:id - Delete a product from the cart.

For the attendance tracking system:

GET /attendance - Get a list of visits.

POST /attendance - Mark a visit.

GET /attendance/:userId - Get attendance information for a specific user.

PUT /attendance/:id - Update attendance information.

DELETE /attendance/:id - Delete an attendance record.

3. Route documentation:

For each route, provide:

The URL (e.g. /users, /products/:id).

A brief description of the purpose of the route.

4. Error handling:

Implement error handling for typical cases:

Missing or incorrect data entry (not required now).

Resource not found (e.g. user or product with specified ID does not exist).

Internal server errors (e.g. database problems).